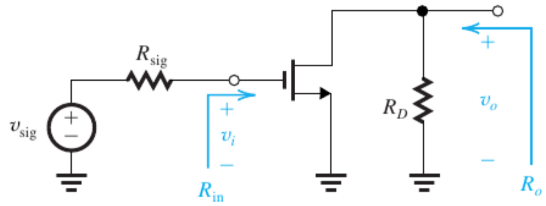
Lecture 7b

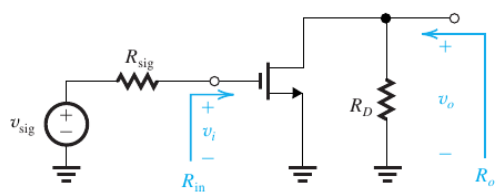
EE-215 Electronic Devices and Circuits

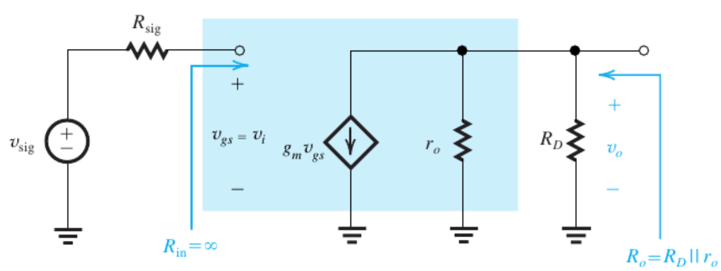
Asst Prof Muhammad Anis Ch

Basic MOSFET Amplifier Configurations

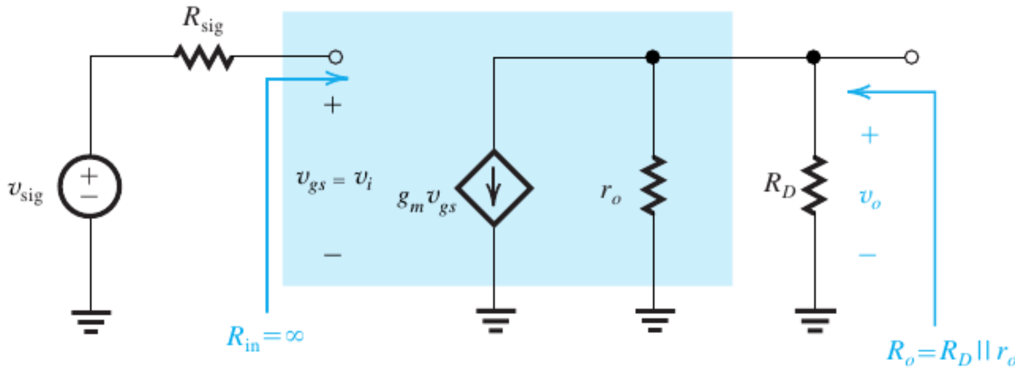
The Common-Source (CS) Amplifier

- is the most widely used MOS amplifier configuration
    - 

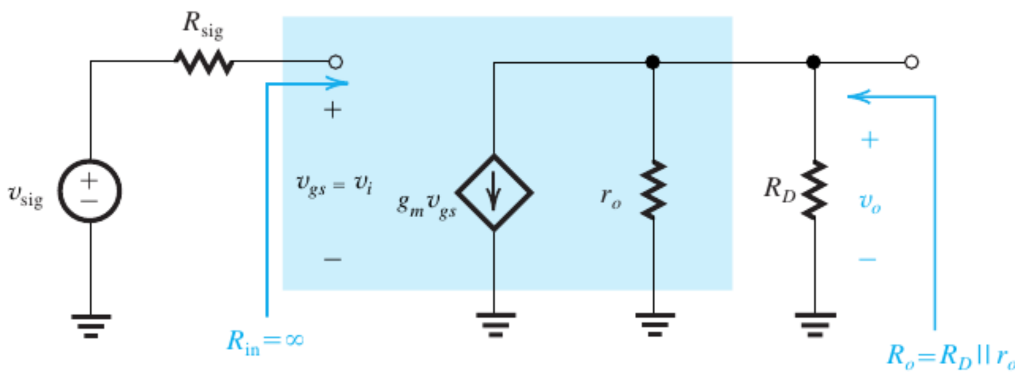
(a)
    - Note that, here the biasing arrangement is not shown (the circuit in fig is the ac equivalent circuit)
    - the signal source has an internal resistance of R_{sig} and open circuit voltage of v_{sig}
    - let's analyze this circuit to determine R_{in} , A_{vo}, R_o and G_v
    - Note that R_D is part of the amplifier circuit, and when a load resistance R_L is connected
        - to the amplifier output, it appears in parallel with R_D
- replacing MOSFET with its hybrid-π model
    - 

(a)


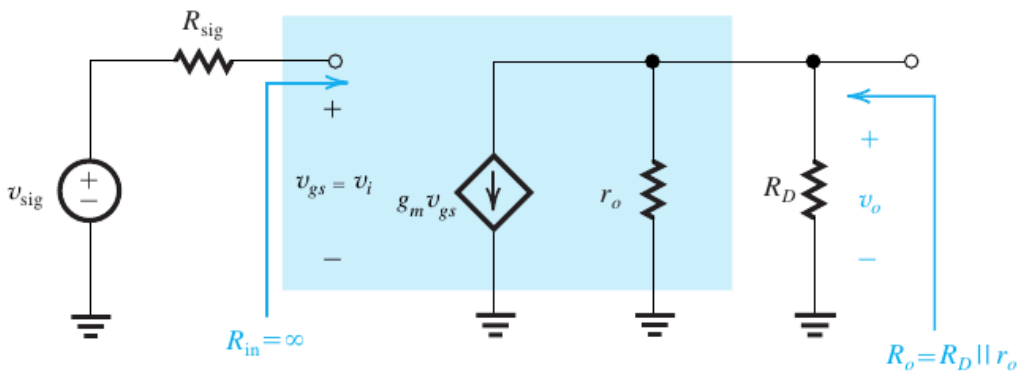
(b)
Figure 5.45 **(a)** Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig}. The bias circuit is omitted. **(b)** The common-source amplifier with the MOSFET replaced with its hybrid-π model.



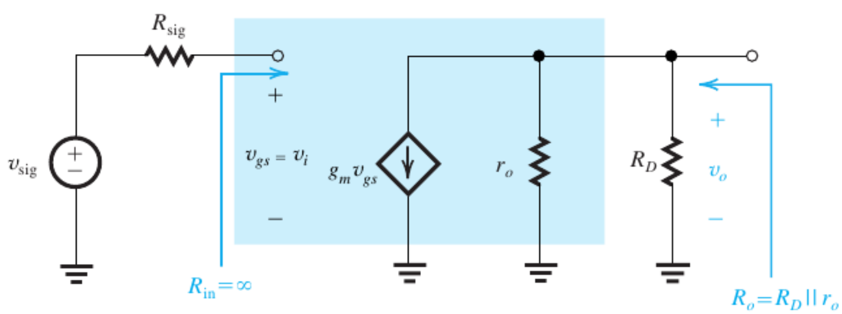
- (b)
  - this equivalent circuit can be used to determine the characteristic parameters (R_{in}, A_{vo} , R_o) of the amplifier
  - from the fig
    - as gate current is zero $= \Rightarrow$ input resistance $= R_{in} = \infty$
  - by KCL $= \Rightarrow g_m v_{gs} + \frac{v_o}{r_o \| R_D} = 0$
    - $v_o = -g_m v_{gs}\left(r_o \parallel R_D\right) = \Rightarrow \frac{v_o}{v_{gs}} = -g_m\left(r_o \parallel R_D\right)$



- (b)
  - $\frac{v_o}{v_{gs}} = -g_m\left(r_o \parallel R_D\right)$
  - here $v_i = v_{gs} = \Rightarrow A_{vo} = . \left.\frac{v_o}{v_i}\right|_{R_L=\infty} = . \left.\frac{v_o}{v_{gs}}\right|_{R_L=\infty} = -g_m\left(r_o \parallel R_D\right)$
  - if the channel length modulation can be ignored $r_o \to \infty$ ($r_o > R_D$)
    - $= \Rightarrow A_{vo} \approx -g_m R_D$

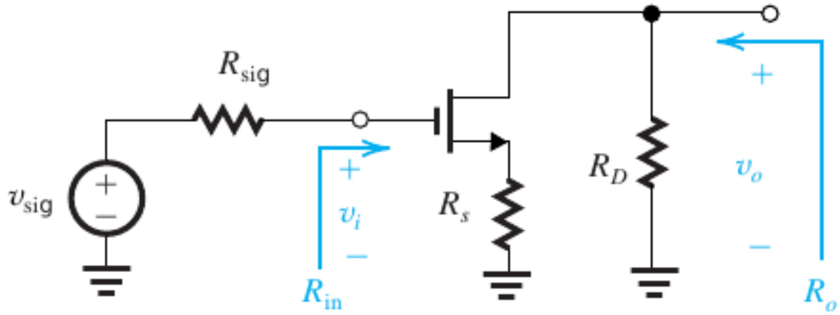


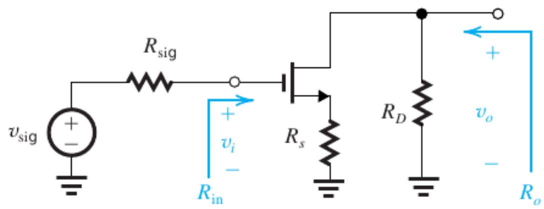
- (b)
    - Now the output resistance R_o is the resistance seen looking into the output terminal with v_i set to zero
        - $v_i = 0 = \Rightarrow v_{gs} = 0 = \Rightarrow g_m v_{gs} = 0$
        - $= \Rightarrow R_o = R_D \parallel r_o$
    - thus for CS amplfier
        - $R_{in} = \infty$, $R_o = R_D \parallel r_o$, $A_{vo} = -g_m \left(R_D \parallel r_o\right)$

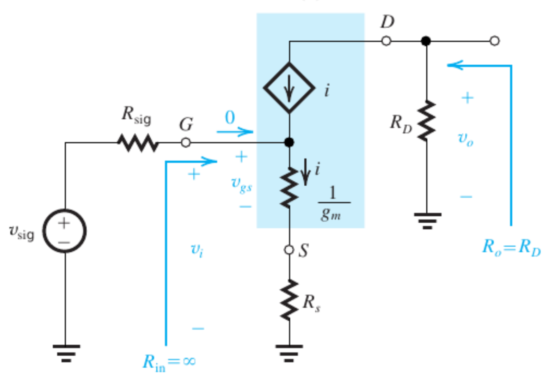


- (b)
    - now the overall voltage gain G_v can be determined as follows
        - $G_v = \frac{v_o}{v_{sig}}$
    - As $R_{in} = \infty = \Rightarrow v_{sig} = v_i = v_{gs}$
    - $= \Rightarrow G_v = \frac{v_o}{v_{sig}} = \frac{v_o}{v_i} = A_v$
    - Note that A_v , G_v are calculated with load resistor R_L connected$= \Rightarrow$ $v_o = \left(-g_m v_{gs}\right)\left(R_D \parallel r_o \parallel R_L\right)$
        - $\frac{v_o}{v_{gs}} = -g_m \left(R_D \parallel r_o \parallel R_L\right) = \frac{v_o}{v_i} = G_v = A_v \because v_i = v_{gs}$

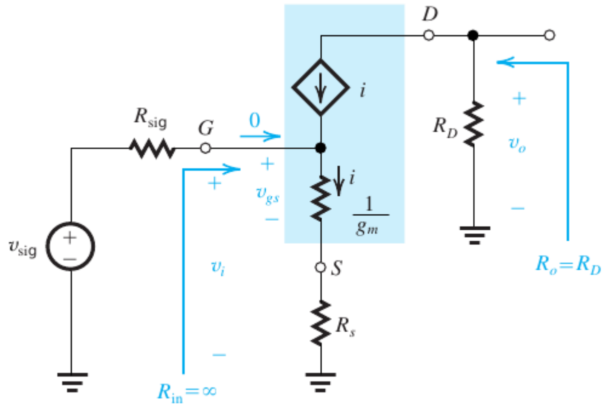
The Common-Source (CS) Amplifier with a Source Resistance

- 

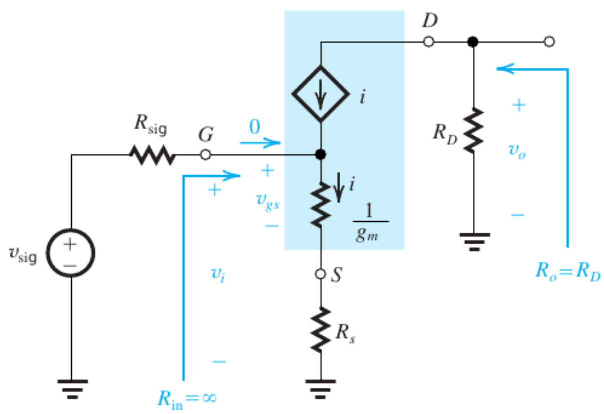
(a)
    - here we have placed a resistance R_s in the source lead of the common-source amplifier
    - Assuming $\lambda = 0$ i.e. ignoring channel length modulation
- the small-signal equivalent circuit can be drawn as (for $\lambda = 0$)
    - 

(a)


(b)
    - here T-model is preferred over the π model, because it makes the analysis somewhat simple
    - In general, whenever a resistance is connected in the source lead, the T model is preferred.

- 

  (b)

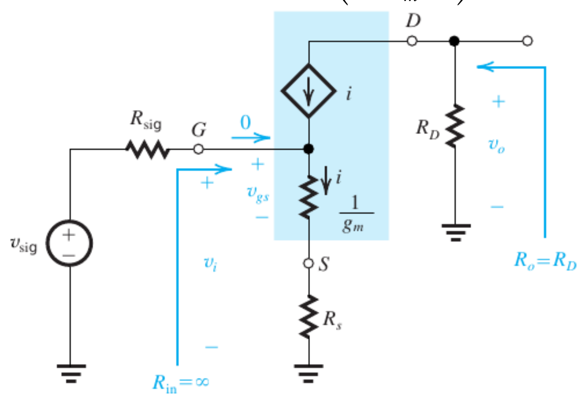
  - As a consequence of the T-model, R_s appears in series with the resistance $1/g_m$
  - as gate current is zero $= \Rightarrow R_{in} = \infty = \Rightarrow v_i = v_{sig}$
  - here only a fraction of v_i appears between gate and source as v_{gs}
  - by voltage divider $= \Rightarrow v_{gs} = \frac{(1/g_m)}{(1/g_m)+R_s} v_i = \frac{1}{1+g_m R_s} v_i$
- 

  (b)

  - $v_{gs} = \frac{(1/g_m)}{(1/g_m)+R_s} v_i = \frac{1}{1+g_m R_s} v_i$
    - thus the value of R_s can be used to control the magnitude of the signal v_{gs}
    - and thereby ensure that v_{gs} doesnot become too large to cause
    - unacceptably high non-linear distortion
  - Another benefit is that R_s extends the useful bandwidth of the amplifier
- by ohm's law,
  - $v_o = -iR_D$
    - As $1/g_m$ and R_s are in series, the current i in the source lead

can be given as

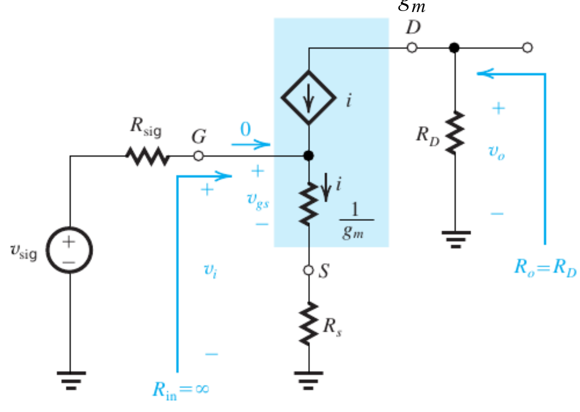
- $i = \frac{v_i}{1/g_m + R_s} = \left(\frac{g_m}{1+g_m R_s}\right) v_i$
- $= \Rightarrow v_o = -iR_D = -\left(\frac{g_m}{1+g_m R_s}\right) v_i R_D$



(b)

- the voltage gain, $A_{vo} = .\ \frac{v_o}{v_i}\Big|_{R_L=\infty} = -\left(\frac{g_m}{1+g_m R_s}\right) R_D = \frac{-g_m R_D}{1+g_m R_s}$
- $= \Rightarrow$ the resistance R_s reduces the voltage gain by the factor $\left(1 + g_m R_s\right)$

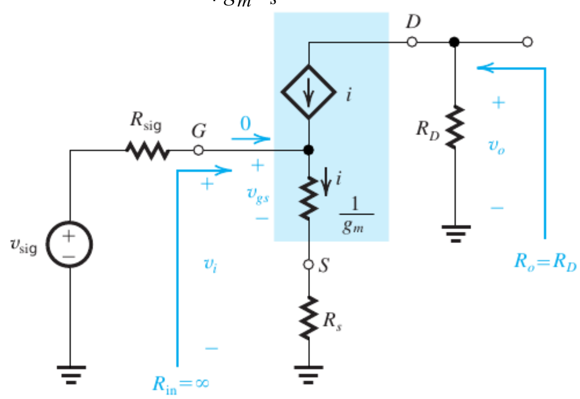
- $A_{vo} = -\frac{g_m R_D}{1+g_m R_s}$
  - divide numerator and denominator by g_m on R.H.S
  - $A_{vo} = -\frac{R_D}{\frac{1}{g_m} + R_s}$
    - $= \Rightarrow$ the voltage gain between gate and the drain is equal to the ratio of the total resistance in the drain (R_D) to the total resistance in the source ($\frac{1}{g_m} + R_s$)



(b)

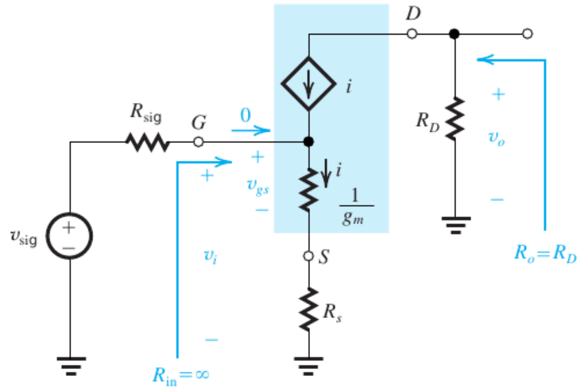
  - i.e. $Voltage\ gain\ from\ gate\ to\ drain = -\frac{Total\ Resistance\ in\ drain}{Total\ Resistance\ in\ source} = -\frac{R_D}{\frac{1}{g_m} + R_s}$

- Note that setting $R_s = 0$ gives A_{vo} of the CS amp$= -\frac{R_D}{(1/g_m)}$

- the output resistance R_o can be determined by placing a test source at the drain
  - and by setting $v_i = 0 = \Rightarrow v_{gs} = 0 = \Rightarrow i = g_m v_{gs} = 0$
  - $i = 0 = \Rightarrow R_o = R_D$
    - now when R_L is connected at the output, A_v can be given as
      - $A_v = A_{vo} \frac{R_L}{R_L + R_o}$
    - As $A_{vo} = -\frac{g_m R_D}{1 + g_m R_s}$, $R_o = R_D$
    - 

(b)
  - $= \Rightarrow A_v = A_{vo} \frac{R_L}{R_L + R_o} = \left(-\frac{g_m R_D}{1 + g_m R_s}\right) \frac{R_L}{R_L + R_D}$

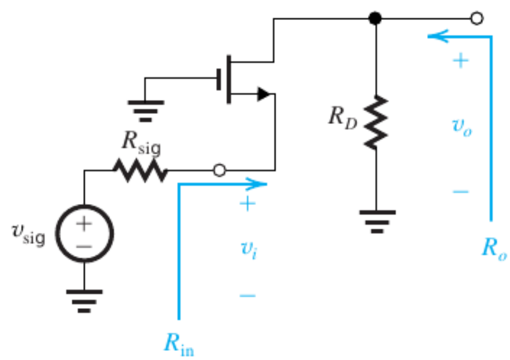
- $A_v = \left(-\frac{g_m R_D}{1 + g_m R_s}\right) \frac{R_L}{R_L + R_D} = \left(-\frac{g_m}{1 + g_m R_s}\right) \frac{R_D R_L}{R_L + R_D}$
  - $A_v = \left(-\frac{g_m}{1 + g_m R_s}\right) \left(\frac{1}{\frac{1}{R_D} + \frac{1}{R_L}}\right) = -\frac{g_m (R_D \| R_L)}{1 + g_m R_s}$
  - $A_v = -\frac{R_D \| R_L}{(1/g_m) + R_s}$
    - Note that $-A_v$ is the ratio of the total resistance in drain to the total resistance in source
    - As R_{in} is infinite $= \Rightarrow v_i = v_{sig}$

- 

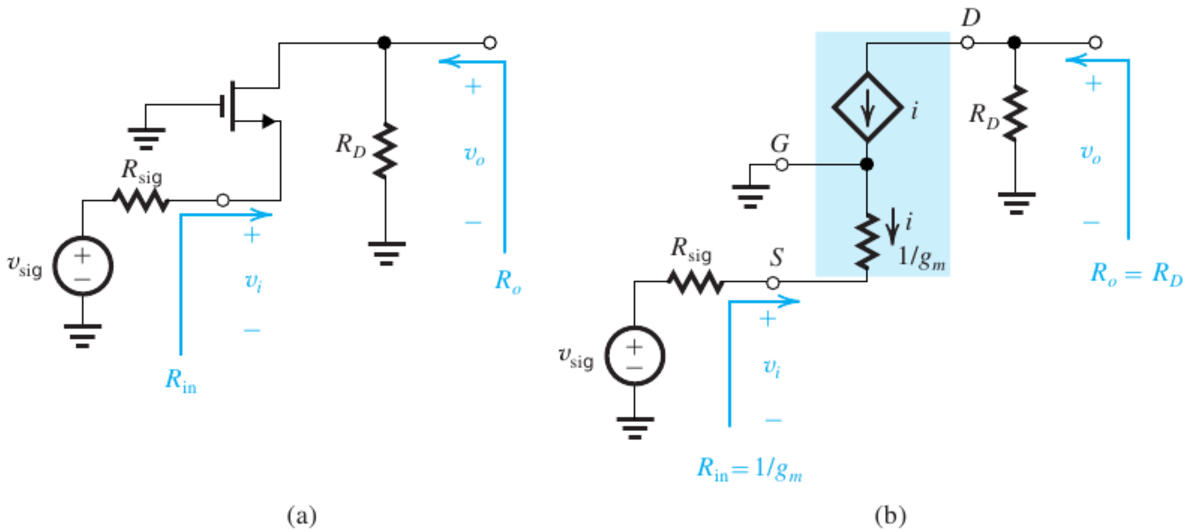
(b)

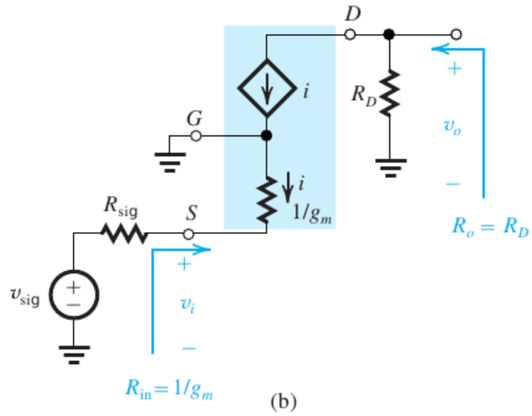
- and the overall voltage gain is $G_v = \frac{v_o}{v_{sig}} = \frac{v_o}{v_i} = A_v$

The Common-Gate (CG) Amplifier

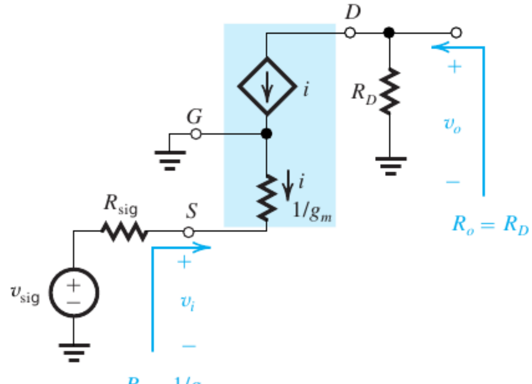
- here common-gate amplifier is shown
  - Note that the biasing circuit is not shown in figure
    - the amplifier is fed with a signal source characterized by v_{sig} and R_{sig}
    - Note that R_{sig} appears in series with the source
      - $= \Rightarrow$it is more convenient to use the T model rather than the π model
    - 

(a)

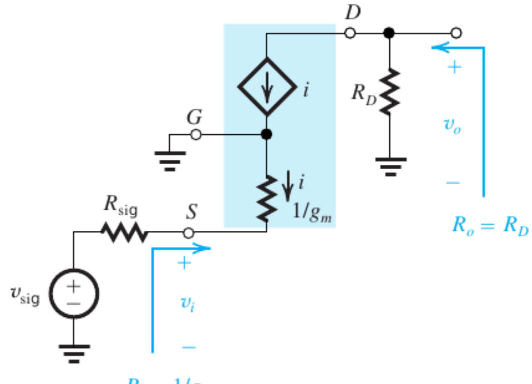


- **Figure 5.48** (**a**) Common-gate (CG) amplifier with bias arrangement omitted. (**b**) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.
    - here channel length modulation is ignored
- the input resistance R_{in} can be given as $R_{in} = \frac{1}{g_m}$
    - as R_{in} is the resistance looking into the source and the gate is grounded
    - typically $1/g_m$ is a few hundred ohms
        - $= \Rightarrow$ the CG amplifier has a low input resistance
        - by ohm's law
            - $v_o = (-i)R_D$,
            - where i is also the source current
            - i.e. $i = \frac{0 - v_i}{1/g_m} = -v_i g_m$
        - 

    - $= \Rightarrow v_o = v_i g_m R_D = \Rightarrow A_{vo} = \frac{v_o}{v_i} = g_m R_D = \frac{R_D}{1/g_m}$
- The output resistancee of the CG amplifier is
    - $R_o = R_D$
    - the gain A_v is

- $A_v = \frac{v_o}{v_i} = g_m\left(R_D \parallel R_L\right)$
  - $\because A_{vo} = g_m R_D$ and as R_L is in parallel with R_D
- 

(b)

- To determine the overall gain G_v
  - first by voltage divider
    - $v_i = \frac{(1/g_m)}{(1/g_m)+R_{sig}} v_{sig} = \frac{R_{in}}{R_{in}+R_{sig}} v_{sig}$
    - $G_v = \frac{v_o}{v_{sig}} = \frac{v_o}{v_i}\frac{v_i}{v_{sig}}$
      - $G_v = \frac{v_o}{v_i}\frac{v_i}{v_{sig}} = A_v \frac{v_i}{v_{sig}} = A_v \frac{(1/g_m)}{(1/g_m)+R_{sig}}$
    - $\because A_v = g_m\left(R_D \parallel R_L\right)$
      - $G_v = g_m\left(R_D \parallel R_L\right)\frac{(1/g_m)}{(1/g_m)+R_{sig}}$
      - $G_v = \frac{R_D \| R_L}{(1/g_m)+R_{sig}}$
    - 

(b)

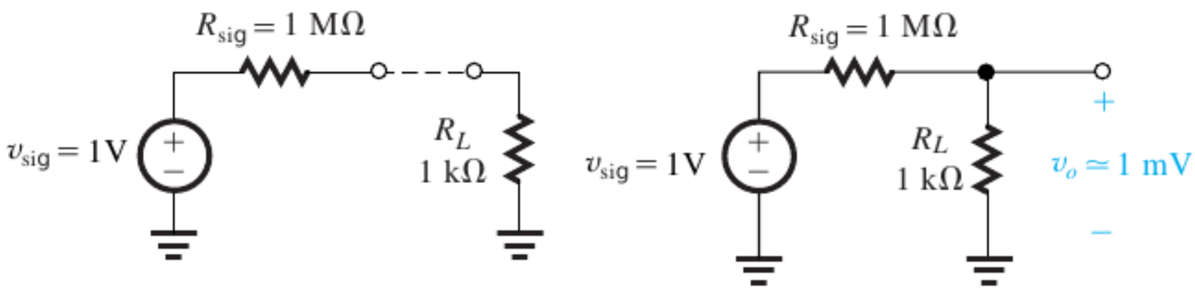
- $G_v = \frac{R_D \| R_L}{(1/g_m)+R_{sig}}$
  - Note that the overall voltage gain is simply the ratio of
    - the total resistance in the drain circuit
    - to the total resistance in the source circuit

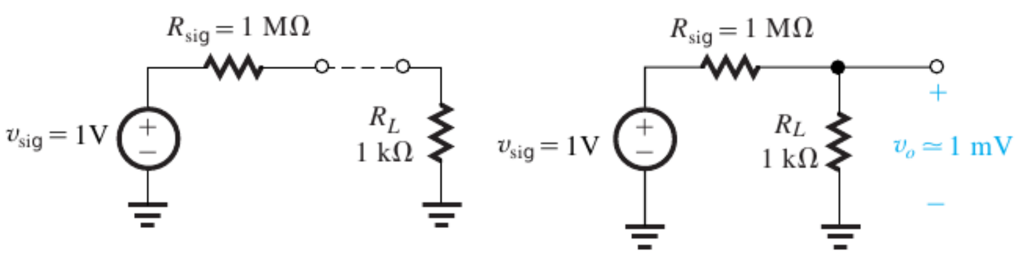
- CG amplifier has excellent frequency response and can be used to
  - amplify high-frequency signals that originate from sources
  - with relatively low resistances

The Common-Drain Amplifier or Source Follower

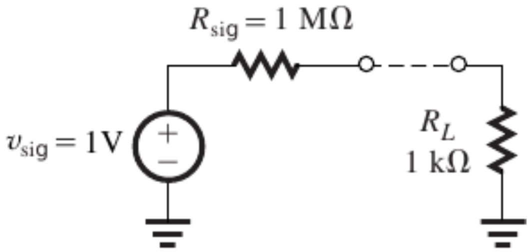
- The common-drain amplifier is more commonly known as the source follower
  - A common application of a CD amplifier is as a voltage buffer

The need for Voltage Buffers

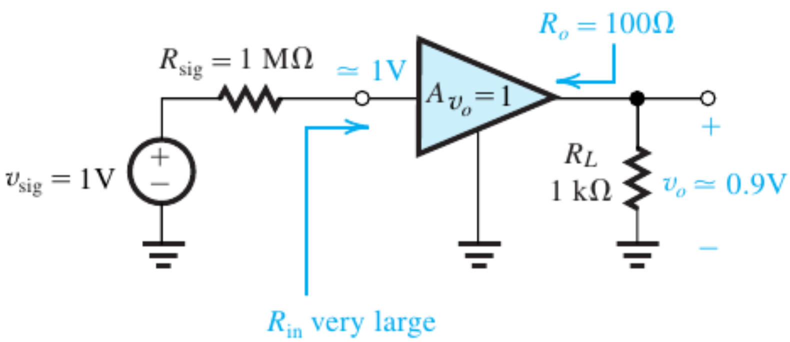
- 

(a) (b)
  - consider the situation shown in fig
    - here a signal source with open-circuit voltage of $v_{sig} = 1V\ peak$
      - and internal resistance of $1M\Omega$ is to be connected to a $1k\Omega$ load resistance
- 

(a) (b)
  - connecting the source to the load directly will result in
    - severe attenuation of the signal
      - i.e. the voltage appearing accross the load is
      - $v_o = \frac{R_L}{R_L+R_{sig}} v_{sig} = \frac{1k}{1k+1M} v_{sig}$
    - As $v_{sig} = 1V = \Rightarrow v_o = \frac{1k}{1k+1M} v_{sig} = 0.999mV \approx 1mV$
    - thus for $v_{sig} = 1V\ peak$, output voltage is only $v_o = 1mV\ peak$

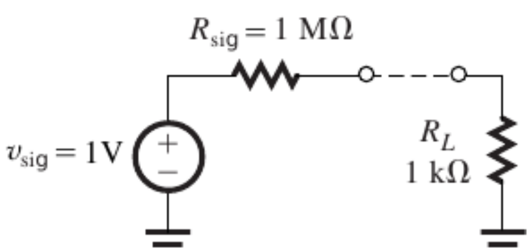


(a)

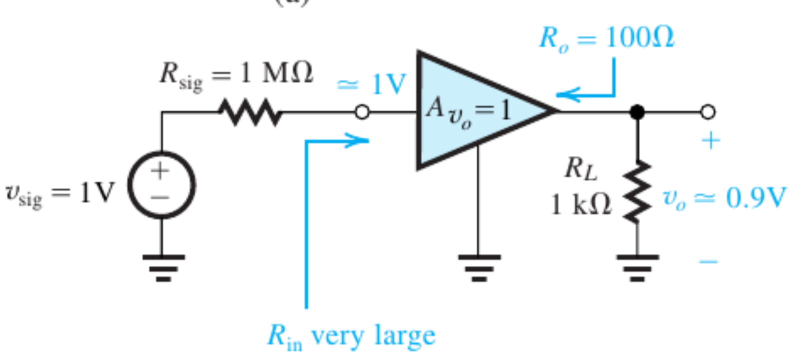


(c)

- now suppose, we have an amplifier as shown in fig
    - this amplifier is used to connect the source to the load
        - this amplifier has a unity voltage gain
            - (here our signal is already of sufficient strength and increase in its amplitude is not required)



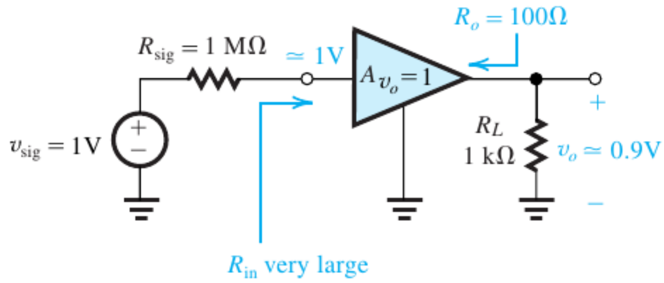
(a)



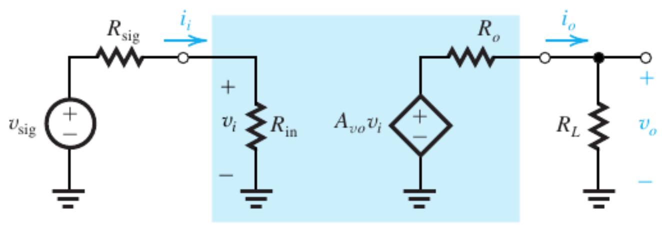
(c)

- this amplifier has a very large input resistance (R_{in} is very large)

- $= \Rightarrow v_i \approx v_{sig}$
  - the amplifier has an output resistance of
    - $R_o = 100\Omega$



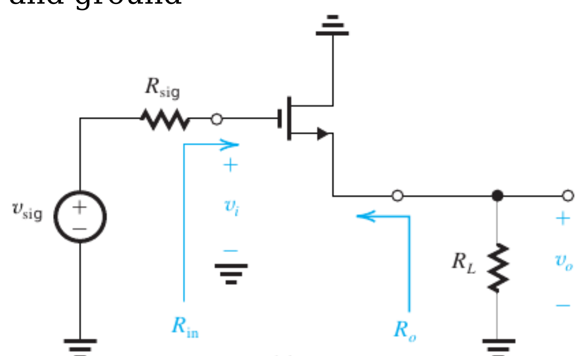
(c)



(b)

- here $R_o = 100\Omega$, $R_L = 1k\Omega$, $v_i \approx v_{sig}$, $A_{vo} = 1$
  - by voltage divider
    - $v_o = \frac{R_L}{R_L+R_o} A_{vo} v_i = \frac{1k}{1k+100} \left(1\right) v_{sig} = 0.9V$
    - $= \Rightarrow 90\%$ of the input signal appears at the output
    - such an amplifier circuit with $A_{vo} = 1$, $R_{in} \to \infty$ and $R_{out} \to 0$ is called a voltage buffer

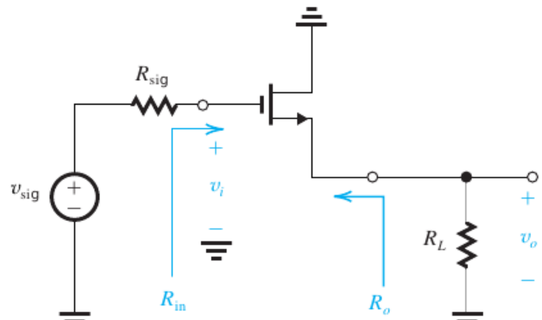
The Common-Drain Amplifier or Source Follower

- The source follower can be used to implement the unity gain buffer amplifier
  - the CD amplifier with the bias circuit omitted is shown here
    - a signal generator (v_{sig}, R_{sig}) is applied between the gate and ground
    - a load resistance R_L is connected between the source terminal and ground
    - 

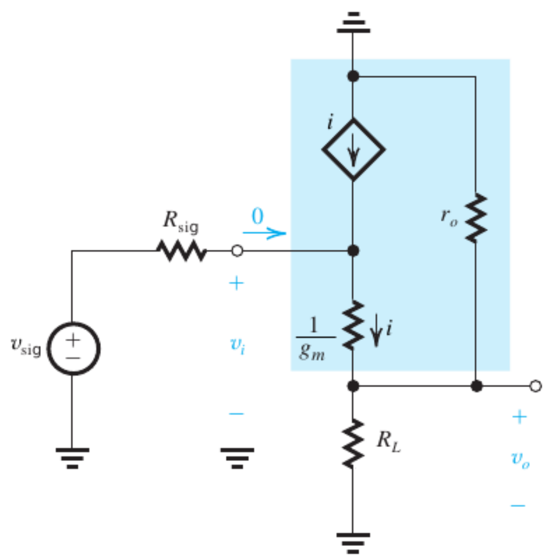
(a)

- As R_L is connected to the source terminal, it is more convenient to use the T-model rather that the hybrid-π model

- here r_o appears in parallel with R_L. so channel length modulation effect can easily be incorporated.

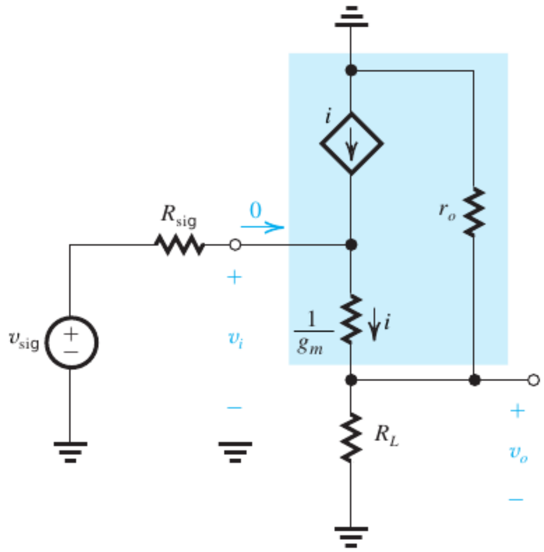


(a)

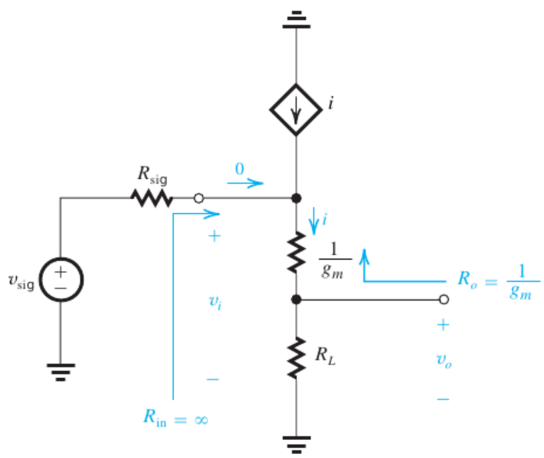


(b)

- if we ignore channel length modulation $= \Rightarrow r_o = \infty$

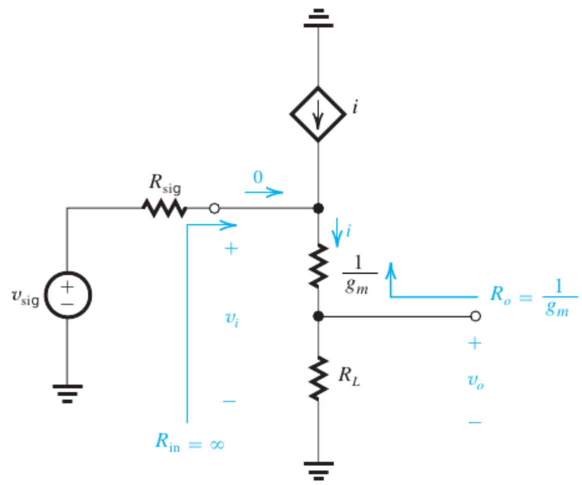


- (b)



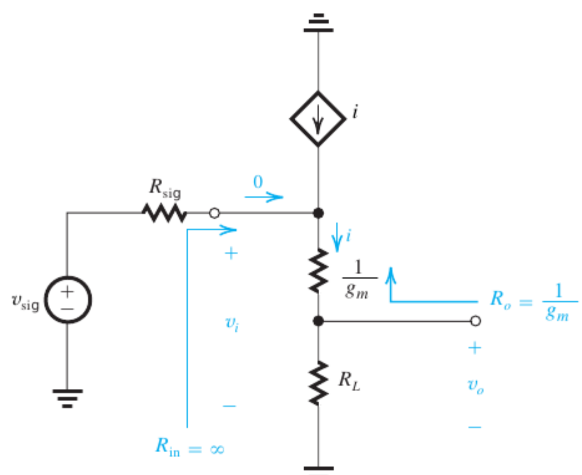
(c)

- from figure
  - $R_{in} = \infty$
    - as no current flows in to the gate terminal
    - $= \Rightarrow v_i = v_{sig}$
    - by voltage divider
      - $\frac{v_o}{v_i} = \frac{R_L}{R_L + (1/g_m)} = A_v$
    - and A_{vo} can be given as
      - $A_{vo} = . \left.\frac{v_o}{v_i}\right|_{R_L=\infty} = . \left.\frac{R_L}{R_L + (1/g_m)}\right|_{R_L=\infty}$

- 

(c)

  - $A_{vo} = . \frac{1}{1+\frac{1}{g_m}\frac{1}{R_L}}\Big|_{R_L=\infty} = \frac{1}{1+0} = 1$

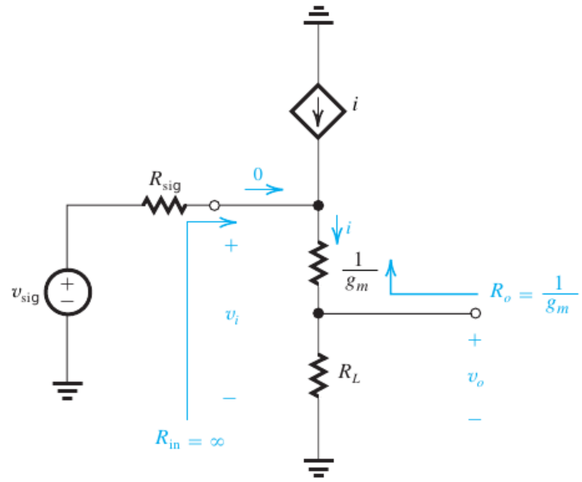
- the output resistance R_o is determined by setting $v_i = 0$ and $R_L = \infty$
  - $= \Rightarrow v_i = 0 = \Rightarrow$ the gate terminal is grounded
    - $= \Rightarrow R_o = \frac{1}{g_m}$
    - $G_v = \frac{v_o}{v_{sig}} = \frac{v_o}{v_i} \because v_i = v_{sig}$ because of infinite R_{in}
      - $G_v = \frac{v_o}{v_i} = A_v = \frac{R_L}{R_L+(1/g_m)}$
      - thus G_v will be lower than unity
    - 

(c)

  - however, as $1/g_m$ is usually a small value
    - $= \Rightarrow$ voltage gain will be close to unity i.e $G_v = A_v \approx 1$

- $G_v = \frac{v_o}{v_i} = A_v \approx 1$ and $A_{vo} = 1$
  - $= \Rightarrow$the voltage at the source terminal will follow that at the input,

hence the name source follower



(c)

- thus the source follower has $R_{in} = \infty$, $R_o = \frac{1}{g_m}$ (a small value) and $A_{vo} = 1, A_v = G_v \approx 1$
- = $\Rightarrow$ the source follower is ideally suited for implementing the unity gain voltage buffer.

Table 5.4 Characteristics of MOSFET Amplifiers

	Characteristics[a, b]				
Amplifier type	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 5.45)	∞	$-g_m R_D$	R_D	$-g_m(R_D \parallel R_L)$	$-g_m(R_D \parallel R_L)$
Common source with R_s (Fig. 5.47)	∞	$-\dfrac{g_m R_D}{1 + g_m R_s}$	R_D	$\dfrac{-g_m(R_D \parallel R_L)}{1 + g_m R_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m + R_s}$	$-\dfrac{g_m(R_D \parallel R_L)}{1 + g_m R_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m + R_s}$
Common gate (Fig. 5.48)	$\dfrac{1}{g_m}$	$g_m R_D$	R_D	$g_m(R_D \parallel R_L)$	$\dfrac{R_D \parallel R_L}{R_{sig} + 1/g_m}$
Source follower (Fig. 5.50)	∞	1	$\dfrac{1}{g_m}$	$\dfrac{R_L}{R_L + 1/g_m}$	$\dfrac{R_L}{R_L + 1/g_m}$

[a] For the interpretation of R_{in}, A_{vo}, and R_o, refer to Fig. 5.44(b).

[b] The MOSFET output resistance r_o has been neglected, as is permitted in the discrete-circuit amplifiers studied in this chapter. For IC amplifiers, r_o must always be taken into account.